THOSE DATA ARE FROM 8 YEARS AGO GSE42826 Active TB vs healthy controls, whole blood Illumina HumanHT-12 V4.0 (GPL10558) 52 controls, 11 TB 2

GSE42830 Active TB vs healthy controls, whole blood Illumina HumanHT-12 V4.0 (GPL10558) 38 controls, 16 TB 2
GSE34608 Validation: TB vs controls, whole blood Agilent 4×44K (GPL6480)

More recent TB transcriptomic datasets and technologies are available

Your concern is valid: GSE42826/42830/34608 are older microarray cohorts (2010–2013). Newer TB datasets use **larger cohorts** and **RNA-seq/single-cell** methods.

Newer bulk blood transcriptome datasets (TB vs LTBI/controls)

Recent multicohort and biomarker studies draw mainly on these GEO series, which you can reuse instead of GSE42826/30:

Whole-blood microarrays (Illumina GPL10558 etc.)

Use case	Example GEO series (year in source paper)	Notes	Sources
Adult ATB vs LTBI/controls, large cohorts	**GSE37250, GSE39939, GSE39940, GSE19491, GSE28623, GSE40553, GSE101705, GSE112104**	11 cohorts, 2,758 patients across 8 countries; used to derive S100A12/S100A8 LTBI–ATB classifier (Frontiers Microbiol 2025) (Jiang et al., 2025)	(Jiang et al., 2025; Wang et al., 2023; Gong et al., 2021; Natarajan et al., 2022)
Host-response signatures, culture-confirmed ATB	**GSE19491, GSE28623, GSE29536, GSE34608, GSE37250, GSE39939, GSE39940, GSE41055**	Compared 16 diagnostic signatures (PLoS Med 2019) (Warsinske et al., 2019)	(Warsinske et al., 2019; Natarajan et al., 2022)
Pediatric LTBI vs ATB	**GSE39939, GSE39940**	HIV-negative children; 265 DEGs distinguishing LTBI vs ATB (Shao et al., 2021)	(Shao et al., 2021; Jiang et al., 2025)

These series are repeatedly re-analyzed in 2021–2025 work, so they are “current” in usage even if generated earlier.

RNA-seq and other modern data

- **Whole-blood RNA-seq treatment cohorts**: 79 TB patients with serial RNA-seq at diagnosis, weeks 1, 4, 24; used to quantify M. tuberculosis rRNA dynamics (Lopez-Exposito et al., 2025).
- **Serum exosome small-RNA-seq**: distinct miRNA panels for LTBI vs ATB vs healthy (Lyu et al., 2019).
- **Metagenomic lung microbiome (BALF)**: shotgun mNGS profiles in untreated, treated, and cured pulmonary TB (Xiao et al., 2022).
- **Single-cell lung datasets**: scRNA-seq of M. tuberculosis–infected macrophages in mouse and human lungs, with multi-modal measurements (Pisu et al., 2021; Pisu & Russell, 2023).

Example: where to start

If you want something more modern than GSE42826/30 but still blood-based and large:

- For **ATB vs LTBI**: start with **GSE37250** (training) and validate on **GSE19491** or pediatric **GSE39939/39940** (Jiang et al., 2025; Warsinske et al., 2019; Natarajan et al., 2022).
- For **RNA-seq & treatment response**: use the 79-patient whole-blood RNA-seq cohort (Lopez-Exposito et al., 2025).

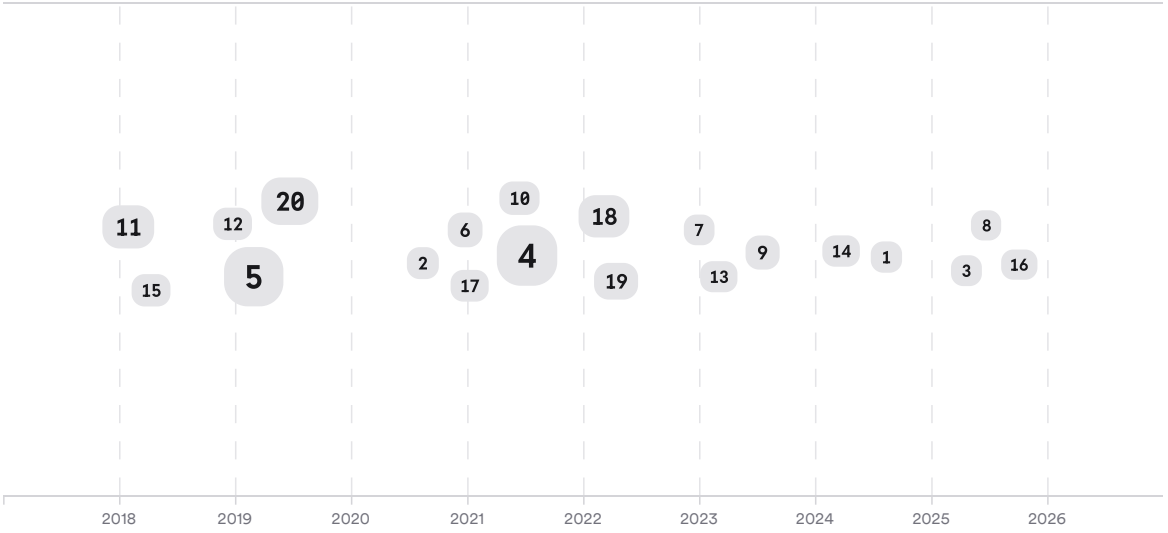

FIGURE 1 Shift from small microarrays to large cohorts and RNA-seq.

Conclusion

Instead of relying on the 2010–2013 GSE42826/42830/34608 sets, you can use larger multicohort Illumina blood microarrays (e.g., GSE37250, GSE39939/40, GSE19491) or newer RNA-seq and single-cell datasets that are actively used in 2021–2025 TB biomarker research.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

Jiang, F., Liu, Y., Li, L., Ni, R., An, Y., Li, Y., Zhang, L., & Gong, W. (2025). Genome-wide expression in human whole blood for diagnosis of latent tuberculosis infection: a multicohort research. *Frontiers in Microbiology*, 16. https://doi.org/10.3389/fmicb.2025.1584360

Pisu, D., Huang, L., Narang, V., Theriault, M., Lê-Bury, G., Lee, B., Lakudzala, A., Mzinza, D., Mhango, D., Mitini-Nkhoma, S., Jambo, K., Singhal, A., Mwandumba, H., & Russell, D. (2021). Single cell analysis of M. tuberculosis phenotype and macrophage lineages in the infected lung. *The Journal of Experimental Medicine*, 218. https://doi.org/10.1084/jem.20210615

Warsinske, H., Vashisht, R., & Khatri, P. (2019). Host-response-based gene signatures for tuberculosis diagnosis: A systematic comparison of 16 signatures. *PLoS Medicine*, 16. https://doi.org/10.1371/journal.pmed.1002786

Wang, C., Hua, J., He, X., Chen, L., & Lv, S. (2023). A diagnostic model for distinguishing between active tuberculosis and latent tuberculosis infection based on the blood expression profiles of autophagy-related genes. *Therapeutic Advances in Respiratory Disease*, 17. https://doi.org/10.1177/17534666231217798

Lopez-Exposito, P., Seral, J., Ferrer, S., Mendoza, A., Gordaliza, P., & Vaquero, J. (2025). Exploring Mycobacterium tuberculosis rRNA Transcriptomic Signatures as Response to Anti-TB Treatment in Whole Blood RNA-seq Data. *bioRxiv*. https://doi.org/10.1101/2025.06.16.659901

Gong, Z., Gu, Y., Xiong, K., Niu, J., Zheng, R., Su, B., Fan, L., & Xie, J. (2021). The Evaluation and Validation of Blood-Derived Novel Biomarkers for Precise and Rapid Diagnosis of Tuberculosis in Areas With High-TB Burden. *Frontiers in Microbiology*, 12. https://doi.org/10.3389/fmicb.2021.650567

Pisu, D., & Russell, D. (2023). Protocol for multi-modal single-cell RNA sequencing on M. tuberculosis-infected mouse lungs. *STAR Protocols*, 4. https://doi.org/10.1016/j.xpro.2023.102102

Shao, M., Wu, F., Zhang, J., Dong, J., Zhang, H., Liu, X., Liang, S., Wu, J., Zhang, L., Zhang, C., & Zhang, W. (2021). Screening of potential biomarkers for distinguishing between latent and active tuberculosis in children using bioinformatics analysis. *Medicine*, 100. https://doi.org/10.1097/md.0000000000023207

Xiao, G., Cai, Z., Guo, Q., Ye, T., Tang, Y., Guan, P., Zhang, J., Ou, M., Fu, X., Ren, L., Yu, M., Wang, Z., Liu, L., Yang, L., & Zhang, G. (2022). Insights into the Unique Lung Microbiota Profile of Pulmonary Tuberculosis Patients Using Metagenomic Next-Generation Sequencing. *Microbiology Spectrum*, 10. https://doi.org/10.1128/spectrum.01901-21

Natarajan, S., Ranganathan, M., Hanna, L., & Tripathy, S. (2022). Transcriptional Profiling and Deriving a Seven-Gene Signature That Discriminates Active and Latent Tuberculosis: An Integrative Bioinformatics Approach. *Genes*, 13. https://doi.org/10.3390/genes13040616

Lyu, L., Zhang, X., Li, C., Yang, T., Wang, J., Pan, L., Jia, H., Li, Z., Sun, Q., Yue, L., Chen, F., & Zhang, Z. (2019). Small RNA Profiles of Serum Exosomes Derived From Individuals With Latent and Active Tuberculosis. *Frontiers in Microbiology*, 10. https://doi.org/10.3389/fmicb.2019.01174